Department of Electrical and Electronic Engineering
Imperial College London

EE3-14

Power Electronics

Course homepage at: http://learn.imperial.ac.uk

Coursework Parts 1 and 2

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1. Introduction

This coursework is done in two parts. In part 1, you will design and simulate a switch mode power supply. In part two, you will build and test it.

- This coursework is done in pairs.
- Please read all of this document before starting the work.

1.1. Aims

- Teach you some practical circuit skills – both design and construction
- Understand the need for a dynamic model of a power converter when designing a control system to adequately regulate the output.
- Understand the method and limitations of state-space averaging and how it can be used to simplify the control system design of non-linear time invariant systems such as switch-mode power supplies.
- Design a simple controller, analyse its performance and implement it in circuitry
- Choose suitable off-the-shelf components, including a gate driver
- Make a (working!?) power supply on a self-designed PCB

1.2. Assessment

- The coursework makes up 100% of the course.
- Part a) is worth 50% and is assessed by a report (one report per pair).
- Part b) is worth 50% and is assessed in pairs by a final lab viva.
- For your report you are reminded to show your working simulations and provide explanations - do not just present un-interpreted results!
- Mark sheets for part 1 and 2 are given on Blackboard. You should use these to determine the importance of various tasks.
- Report should be no more than 12 pages plus appendices.

1.3. Support

There will be GTAs available to help with this course. They will be available to help with your initial simulation work (which they will mark). You will receive feedback on your initial report before proceeding with your detailed design and layout, which will again be looked at by the GTAs. They will also be in lab to help with circuit construction.

Part 1: System Analysis, Simulation and Design

2. Converter Design

The boost converter is one of the most commonly used switch-mode power supplies (SMPS). It is commonly used in low as well high power applications.

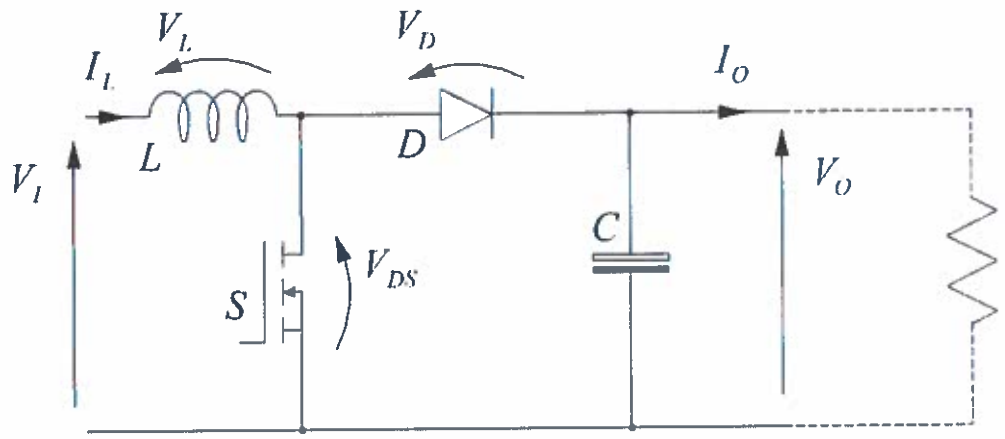


Figure 1: Basic circuit of a boost converter.

Figure 1 shows the basic circuit layout. You will begin by studying the ideal circuit behaviour of a simple boost converter in open loop. You will then improve your simulation model by including real component models and parasitic component effects and study their effects on the converter performance. Following this, measures to improve the efficiency will be introduced. Practical converter implementations commonly include feedback to control the output voltage. This will be introduced and you will design your own compensation network (controller) for your converter.

To help you with your design, hints will be provided along the way.

2.1. Basic Converter Design

First, the basic converter has to be designed to fulfil the desired operational specifications, which are:

- Input voltage: 20 V (±2 V).
- Output voltage: 48 V (±0.5 V ripple).
- Maximum output current: 0.6 A, continuous, resistive loads only.
- Input Current ripple: max 10% at maximum output power.
- A Soft-Start to limit inrush current.
- Switching frequency: <200 kHz.
- Efficiency > 90% at full load.
- Maximum inductor volume of 9 cm^3.

2.1.1. Component Values

Using the following hints and your knowledge of the operational principles of the boost converter, calculate the component values for your converter.

a) Inductor sizing:
   i. Derive the minimum inductor value as a function of input current, input voltage, duty-cycle and switching frequency.
b) In the selection of your inductor (which will be performed later) the peak and RMS inductor currents are both important. Why is this? Also find:
   i. The worst case peak inductor current in steady-state, assuming that your converter operates in continuous mode.
   ii. The worst case RMS inductor current in steady state.
c) Capacitor sizing:
   i. Calculate the minimum capacitor value as a function of duty cycle, switching frequency, load current and voltage ripple.
d) Switch and diode selection:
   i. Find the appropriate value of maximum current and voltage blocking capability for switch and diode and explain the reasons for your choice.

3. Simulation of Ideal Converter in Open Loop

Next build a model of the basic boost converter in PSpice, using an ideal switch (PSpice part name S) in place of the MOSFET and use the **Dbreak** diode component. The inductor and capacitor can be ideal components at this stage. You can create a signal to control the switch using the **VPULSE** component.

You must now run a set of simulations on this converter. These simulations are:

3.1. Simulation sets:

1) Find the efficiency under a range of operating conditions – does this agree with what you expect given the components you are using?
2) Start up: investigate the behaviour of the converter at turn on, *i.e.* when the switch starts to operate assuming the output capacitor and the inductor are in a discharged state. Discuss the following in relation to the transfer function you derived above:
    a. Percentage overshoot of output voltage.
    b. Rise and settling time of output voltage.
    c. Oscillation frequency of output voltage.
3) Change in load: find effects of switching from one extreme load condition to another. NB: You should comment on the output voltage in either condition with regards to the desired value for a fixed duty-cycle. You should examine the response of the converter to stepped load changes between:
    a. 50% Load.
    b. 100% Load.
4) Examine how your converter operates under very low loads.

4. Real Component Choices

Before we build our converter, we must choose some real components that can be purchased easily, from either the OneCall or RS website (links can be found on the EEE store website: http://www.imperial.ac.uk/electrical-engineering/internal/stores/). Once these components have been chosen, we can simulate our converter using realistic models of these components and analyse the converter behaviour before going to the build stage.

Don't forget that a real inductor has some parasitic series resistance, a MOSFET has finite on-state conductance and incurs switching losses. With these things in mind, select appropriate real components for use in your design (MOSFET, capacitor, inductor, diode) in order to meet the specified efficiency.

In your report, keep track of your components and in particular the datasheets for them. Provide a list of your component selection, including from which supplier and their order numbers (you will need them later to order your components).

A note on inductor choice
It is important to use an inductor with a closed-loop core to minimise the amount of magnetic flux escaping the core as this might significantly affect the tracks in the PCB. If no suitable inductor is available off the shelf then you can also wind your own (common practice). In that case you will have to estimate the parasitic series resistance of your inductor. Remember that you should try not to exceed the maximum specified inductor volume.

A note on gate drives
In order to allow the MOSFET to switch quickly with minimal loss, significant charge must be pushed on and off the gate as fast as possible. Dedicated drive chips are available for this purpose – search for "MOSFET gate drive" in RS or OneCall and pick a suitable device.

5. Simulations and Analysis with Real Components

Now that you have chosen your components, you must run a set of simulations which capture the behaviour of the real circuit that you are going to build. In order to do this you will need to have a SPICE model of your chosen diode and MOSFET and you can easily construct a model of your inductor and capacitor.

If you are lucky, PSpice will already include models of the MOSFET and diode you wish to use. If it does not, the manufacturer of the device will provide a text PSpice model description of the component. If you find yourself in this position, you must import the manufacturer model into PSpice, as follows:

Open the PSpice Model Editor.

1) File -> New
2) Model -> Import
3) Save As (Save as "your library name")
4) Copy .OLB file with appropriate graphical model from existing library into the same folder as your library
5) File -> Model Import Wizard [Capture]
6) Leave input and output names as they area & click NEXT
7) Select model name and replace/Associate Symbol
8) Open .OLB library you have previously copied here and select appropriate Capture model
9) Assign pin names (careful they are not always the same order in your Spice model they are suggested in the drop down menu. Check Spice model for accurate allocations).
10) Finish
11) Open Library in Capture and use models. (NB: You may have to restart Capture to load the library properly).

Simulate your converter using real component models (don't forget that your gate drive is not ideal!), again investigating the following:

5.1. Simulations:

1) Find the efficiency under a range of operating conditions – does this agree with what you expect given the components you are using?
2) Start up: investigate the behaviour of the converter at turn on, *i.e.* when the switch starts to operate assuming the output capacitor and the inductor are in a discharged state. Discuss the following in relation to the transfer functions you derived above:
    a. Percentage overshoot of output voltage
    b. Rise and settling time of output voltage
    c. Oscillation frequency of output voltage
3) Change in load: find effects of switching from one extreme load condition to another. NB: You should comment on the output voltage in either condition with regards to the desired value for a fixed duty-cycle. You should examine the response of the converter to stepped load changes between:
    a. 50% Load
    b. 100% Load
4) Examine how your converter operates under very low loads.

6. Synchronous Rectification

You may find that the efficiency target (>90%) of your converter is just about reachable in simulation. In practice, the efficiency is likely to be lower when you have built the circuit and therefore it is prudent at this stage to consider methods of improving the efficiency of the circuit. One possible suggestion is to use a synchronous rectifier where the diode in the converter is replaced by a second MOSFET, sometimes called the sync FET. The idea behind this is that the diode has a relatively fixed voltage drop (around 0.7 V for a pn junction and less for a metal-semiconductor Schottky junction). If a MOSFET is used in place of the diode, the voltage drop across that component can be reduced further. You should now try to simulate a working synchronous boost converter. But be careful:

- If you switch both devices on at once, you will short the input power rail. Think about dead time for switching the devices. Can you see how a half bridge driver could be adapted to this problem?

You should now run some further simulations, this time on the synchronous converter, with real components.

6.1. Simulations

1. Investigate converter behaviour in continuous and discontinuous mode. Does it work as expected? If not, can you make it work?
2. Find the efficiency under a range of operating conditions – does this agree with what you expect given the components you are using?

This completes the open loop simulations and design for your converter. We will next investigate converter behaviour in closed loop.

7. Closed-Loop System

In this section you will design the feedback loop for your boost converter. The task of the compensator is to adjust the duty cycle to provide a constant output voltage regardless of varying load conditions. Here we are concerned with keeping the output voltage constant (within the specified 500 mV) so will choose to regulate the output voltage and thus feed that signal back in the control loop, possibly using a potential divider.

7.1. Static and Dynamic Converter Model

You are familiar with the input to output transfer ratio of this converter in continuous mode, assuming that there are no losses in the circuit, is $V_{out} = V_{in}\ (1/(1-\delta))$. Recalculate this ratio including the non-zero diode drop that is present in a real converter.

To design a controller for the boost converter we require a dynamic model of the boost that captures the transient response of the system to a change in set point.

To do this you must derive state-space models for the converter in continuous mode for each of the two switching states. Using the method of state-space averaging derive a linear state-space model of the converter around the duty cycle set point at full rated load. You may assume an ideal switch, inductor and capacitor in your state-space model and a well regulated voltage supply. See the Additional Notes – State Space Averaging for details on this method.

A boost converter with closed loop control of the output is shown in Figure 2

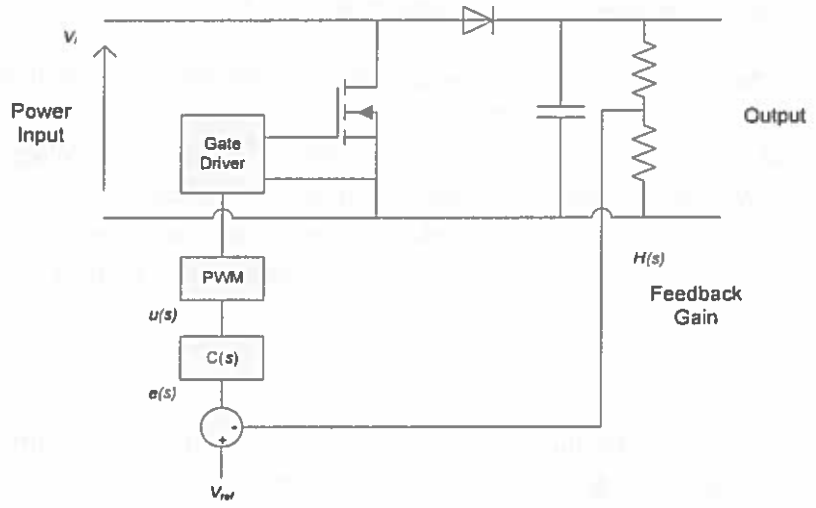


Figure 2 Feedback control of output voltage

The compensator you design ($C(s)$ shown in Figure 3) will be implemented using active filters made with op-amps.

There are different methods available to you for designing the compensation network. The "cookbooks" often distinguish between Type I to Type III compensation for boost SMPSs and academics tend to discuss P, PI, PID, lead, lag etc. You can decide yourself which one you wish to use but you should be able to justify your choice. The example given in the lecture notes is a good start.

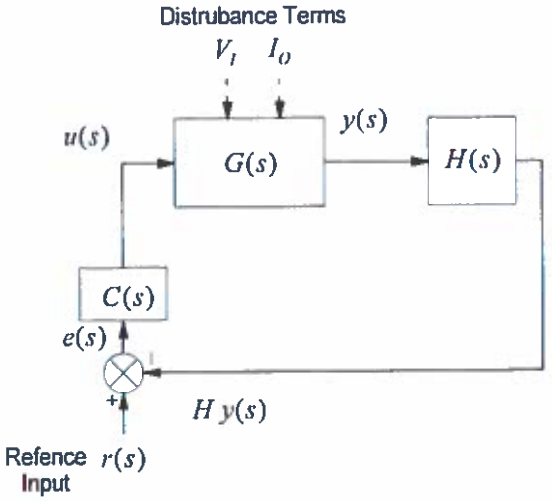


Figure 3 S-domain closed loop boost converter

For further reading please see Chapter 7 of "*Feedback Control Systems*" by Charles L. Phillips and Royce D. Harbor.

State-space averaging has its limitations in that certain assumptions must be made to derive the averaged model. When you have derived your compensator you may wish to check its performance at varying load conditions and supply voltages.

7.2. Compensator Transfer Function Design

The aims of introducing a compensator to your design are as follows:

a. To improve the steady-state output voltage for a given reference input for varying load conditions (i.e. good regulation of output).
b. To soft start the converter, limiting the current inrush when the input voltage is first applied. Discuss what trade-offs, if any, you make to achieve this requirement.
c. Limit the overshoot in output voltage of the converter at turn on. There is a trade of between overshoot and speed of response of the system to a perturbation but trying to limit overshoot to around 10% is a reasonable.

You should now design the transfer function of your compensator, ensuring stability, achieving good load regulation with limited overshoot at turn on. The behaviour of the system can be analysed in Matlab using root locus plots (see the lecture notes).

7.3. Closed loop PSpice Model

Now that you have designed your compensator, you should perform a full time domain simulation of your system in PSpice.

You already have a working model of your converter in open loop, using realistic components. This is the starting point of your closed-loop simulation.

Your compensator transfer function can be constructed in PSpice using parts in the **ABM** library. This contains summations, subtractors, gains etc and a **LAPLACE** block with which you can add the transfer function of the compensator.

The only part of your design which may be missing is the PWM modulator. Up to this point you have simply fed a square wave of variable duty cycle into the converter using the VPULSE component and by varied its parameters. Your closed loop model requires δ to change during run time.

A suitable PWM modulator should take an analogue control signal (the output from your compensator) and be able to turn it into a variable duty cycle signal (the PWM) which will drive the gate.

The PWM modulator can be constructed as shown in *Figure 4*. You will need to add the **ABM** library in order to get the following parts. The **Diff** block subtracter followed an **Etable** voltage-controlled voltage-source provide error amplification. The **Etable** applies the potential difference between the input nodes to a look-up table to generate the output voltage. The look-up table is defined by input-output pairs enclosed in brackets. Interpolation and extrapolation are used to fill out the table.

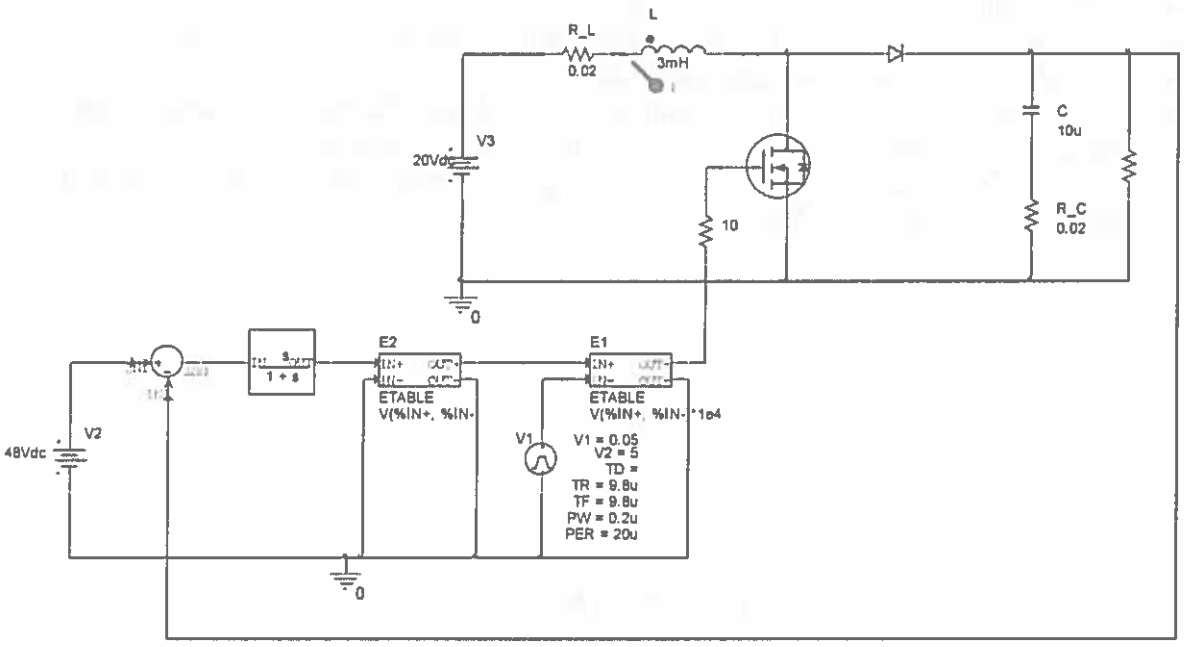


Figure 4 Closed loop control of a boost converter

Define the attributes of the **Etable** component as follows to provide a gain of 1 and also clip the signal to 0.02 to 0.98 volts (this prevents over-driving the modulator):

EXPR=V(%IN+, %IN-)
TABLE=(0,0.02) (1,0.98)

The modulator then compares the amplified error with a triangle wave carrier. Define a pulse signal as follows to obtain a triangle wave at a suitable frequency for your design.

The comparison is done with another **Etable**, voltage-controlled voltage-source which will control the gate drive of your MOSFET. You should set its parameters to:

EXPR=V(%IN+, %IN-)*1e4
TABLE=(-1,0) (1, 15)

This multiplies the input voltage difference by a high gain and then applies a piece-wise linear mapping. The effect is that inputs below -0.1 mV map to 0 V and inputs above 0.1 mV map to 15 V (this will be correct if you are using an n-channel enhancement-mode mosfet but may need modifying if you are using a p-channel device). Between these points linear interpolation is used. The Etable acts as a comparator that gives one of two possible outputs depending on whether the input is positive or negative (and close to zero acts as a real comparator would do).

You may now wish to perform some additional tests to check how well your compensator, designed using an averaged state-space model, performs when the assumptions used in that model are breached.

Report Guidelines

- 12 Pages max limit for main body of text.
- The report should follow the numbering and naming structure of this hand-out.
- Appendices are allowed for supporting material.
- Extra investigation/discussion is encouraged; you will note that there are a large amount of marks allocated under the scientific heading in the marking scheme.
- Your Report should include the names of your group members for the practical part of the course (Worth One Bonus Mark!).

End of tasks for part 1

Part 2: Detailed Design, Build and Test

8. Overview

The second part of the coursework is more flexible than the first: this is intentional so that you have to make decisions without being led step by step by this coursework sheet.

8.1. Design your Compensation Network using Opamps

To implement the compensation network you designed in part 1 you will need to design an opamp implementation of it. Draw a schematic of this circuit. You do not need to simulate the actual opamp implementation of your compensator in PSpice. You should aim to complete the design of this part before you receive your feedback from stage part 1. Any changes necessary to the compensation network should be easy to make once the circuit is designed.

8.2. Design and Build

After you receive feedback from part 1, you should then finish the design. then build your circuit. You can do this either in pairs or by yourself, but the expectation is that you work in pairs. The following bullet points should cover what you need for this part of the activity:

- You already have the basic circuit design for the power converter and have made component choices for the capacitor, inductor, MOSFET(s), diode and gate drive. You do not yet have a circuit design for the PWM module. You should design this and integrate into your design between compensation network (made of opamps) and the gate drive chip.
- Before ordering your components, please have your design checked by one of the GTAs in lab.
- You will then need to order your parts through EE level 1 stores. Please let stores know that this is the for the EE3 Power Electronics course when you place the order.

8.3. Test

Once you have built your circuit, you should test the quality of various performance measures, including:

- Plot of efficiency versus load current 2. plot of output voltage versus load current (to show quality of regulation)
- Response of the output voltage to a step change in input voltage (around the specified operating ranges). You may have to think about how to do this test...
- Response of output voltage to a step change in load. Try going from 50% load to full load to start.

9. FAQ

- Can I buy a PWM chip and a SMPS controller chip from, say TI or Unitrode, instead of designing my own control loop and PWM unit?

  No. You are expected to build these components from more basic components, such as comparators. There is flexibility in the design but we do not want you to buy a complete off the shelf solution!

- How many components should I order?

  Don't go crazy, but at the same time, don't order 1 MOSFET. If you blow it up, you will have to wait for a couple of days until the new one arrives. Order , say, 3 of each fragile device...

- I've seen this incredible specification inductor-like component that is listed as a "choke" on RS/OneCall. Shall I use that?

  No. We can talk about why not in the lectures, but this is going to look more like a wire than an inductor....

- Can I use a digital control loop?

  Yes, you can! But it must be in addition to an analogue based solution... If you have done RTDSP then you may find this an attractive method, but beware of timing of the sampling.....